

Practice math on 2D and 3D Transformation

1. Translate a polygon with co-ordinates A(2,5), B(7,10) and C(10,2) by 3 units in X direction and 4 units in Y direction.
2. A point (4,3) is rotated counter-clockwise by angle of 45. Find the rotation matrix and the resultant point.
3. Apply two scaling to the matrix given below such that $S_{x1}=2$, $S_{y1}=3$, $S_{x2}=4$, $S_{y2}=2$.
$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$
4. Suppose you have the point (2,4), rotate the point by 90 degree then by 180 degree then by 45 degree then by 45 degree then scale it by $S_x=2$ and $S_y=4$.
5. Suppose you have an object (4,5). If we scale the object, then translate to origin then translate back; is it equivalent to translate to origin, scale then translate back? Here the scaling factor is (2,3) and translating factor is (3,2).
6. If we rotate the object (1,3) by 90 degree then scale it ; is it equivalent to scale it then rotate it by 90 degree where the $S_x=2$, $S_y=3$.
7. Rotate the object (5,3,2) by 30 degree about y-axis and then scale it by (2,1.5,3).
8. Suppose you have the point (2,4). Scale the object by 1.5 in X direction and 2.5 in Y direction about an arbitrary center (-2,3).
9. Perform shearing about y-axis on the object (7,5) about a point (3,2). The shearing factor is 2.25.
10. Suppose you have a line $y=3x+2$. Now reflect an object (2,4) about this line and show the result.